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Repeated Low-Level Red Light Therapy for Myopia Control in High Myopia Children and Adolescents

A Randomized Clinical Trial

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Purpose: To assess the effectiveness and safety of repeated low-level red light (RLRL), which is a newly available treatment for myopia control in children and adolescents with high myopia.

Design: Multicenter, randomized, parallel-group, single-blind clinical trial (randomized controlled trial; NCT05184621).

Participants: Between February 2021 and April 2022, 192 children aged 6 to 16 years were enrolled. Each child had at least 1 eye with myopia of cycloplegic spherical equivalent refraction (SER) at least −4.0 diopters (D), astigmatism of ≤2.0 D, anisometropia of ≤3.0 D, and best-corrected visual acuity (BCVA) of 0.2 logarithm of the minimum angle of resolution or better. Follow-up was completed by April 2023.

Methods: Participants were randomly assigned at a 1:1 ratio to intervention (RLRL treatment plus single-vision spectacles) or control (single-vision spectacles) groups. The RLRL treatment was administered for 3 minutes per session, twice daily with a minimum interval of 4 hours, 7 days per week.

Mean Outcome Measures: The primary outcome and key secondary outcome were changes in axial length (AL) and cycloplegic SER measured at baseline and the 12-month follow-up visit. Participants who had at least 1 postrandomization follow-up visit were analyzed for treatment efficacy.

Results: Among 192 randomized participants, 188 (97.91%) were included in the analyses (96 in the RLRL group and 92 in the control group). After 12 months, the adjusted mean change in AL was −0.06 mm (95% confidence interval [CI], −0.10 to −0.02 mm) and 0.34 mm (95% CI, 0.30 to 0.39 mm) in the intervention and control groups, respectively. A total of 48 participants (53.3%) in the intervention group were still experiencing axial shortening >0.05 mm at the 12-month follow-up. The mean SER change after 12 months was 0.11 D (95% CI, 0.02to 0.19 D) and −0.75 D (95% CI, −0.88 to −0.62 D) in the intervention and control groups, respectively.

Conclusions: Repeated low-level red light demonstrates stronger treatment efficacy among those with high myopia, with 53.3% experiencing substantial axial shortening. Repeated low-level red light provides an excellent solution for the management of high myopia progression, a significant challenge in ophthalmology practice.

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Myopia, or nearsightedness, is a prevalent refractive error of the eye characterized by the ability to see close objects clearly while distant objects appear blurred.[1] High myopia not only impairs visual acuity but also substantially raises the risk of sight-threatening complications such as myopic macular degeneration, glaucoma, or retinal detachment.[2]

Despite the growing prevalence and potential vision-threatening consequences of high myopia, there is still a significant gap in evidence-based strategies for its effective control and management. High myopia is often out of range as an indication for orthokeratology.[3,4] Although low-concentration atropine[5-9] or defocus spectacles[10-12] have demonstrated reasonable efficacy on controlling low or moderate myopia in multiple randomized controlled trials, their efficacy in managing high myopia remains unproven.

Repeated low-level red light (RLRL) therapy has emerged as a potential strategy for myopia control.[13-17] A portable desktop device allows for RLRL treatment at home

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proven to be effective in controlling myopia among children with myopia of −1 to −5 diopters (D).[13] The efficacy of RLRL was found to be even greater among individuals with a higher degree of myopia, and a significant proportion of children experienced substantial axial shortening after treatment.

Therefore, we aimed to conduct the first RLRL therapy myopia control trial among high myopia children and adolescents. Additionally, we assessed its safety in this specific group of children.

Methods

Study Design and Setting

This was a prospective, single-blind, parallel-group, multicenter, randomized clinical trial conducted at 6 tertiary hospitals in China. Before commencing the study, all personnel and investigators received sufficient training in the study procedure. The study enrolled its first participant on February 6, 2021, and the last on April 1, 2022. The final follow-up was performed on April 4, 2023. This study upheld the principles of the Declaration of Helsinki, and the Ethics Committees of Shanghai First People's Hospital approved the study protocol. The guardians of the participants provided written informed consent, and the children provided verbal consent. This trial is registered with ClinicalTrials.gov (identifier, NCT05184621).

Eligibility Criteria

This study recruited participants aged 6 to 16 years with ≥1 eye with myopia of cycloplegic spherical equivalent refraction (SER) ≥−4.0 D, astigmatism ≤2.0 D, anisometropia ≤3.0 D, and best-corrected visual acuity (BCVA) of logarithm of the minimum angle of resolution ≥0.2. Children were excluded if they had history or presence of strabismus, ocular trauma, other ocular diseases, systemic diseases, or had received other myopia control treatment, such as atropine therapy, orthokeratology, or Defocus Incorporated Multiple Segments glasses.

Randomization and Masking

A statistician without contacts with the participants generated the randomization sequence using SAS software (version 9.4; SAS Institute, Inc). All eligible participants were randomly assigned to the intervention (RLRL plus single-vision spectacles [SVS]) or control (SVS only) groups at a 1:1 ratio. The randomization process adopted central stratification block randomization with block sizes of 4 or 6. Each eligible participant was assigned a number based on their study entry sequence. Random assignment outcomes were recorded in sealed envelopes with the corresponding sequential numbers written on the covers. Then, an investigator at the main center took custody of the envelopes. Each time an eligible participant's informed consent was acquired, the investigators responsible for group allocation at the 6 centers contacted the investigator holding the envelopes by telephone. The investigator immediately opened the envelopes and provided information on the group allocation of the newly enrolled participants. Because of the nature of the intervention, the participants were aware of their assigned treatment groups. However, outcome assessors, including clinical examination technicians, optometrists, ophthalmologists, and statisticians, were masked to the treatment allocation.

Intervention

According to a routine treatment for myopia, all participants in both groups wore SVS lenses throughout the study. Besides wearing SVS lenses, the intervention group underwent RLRL treatment using a portable desktop device (Eyerising International). The device incorporates a semiconductor laser diode capable of emitting low-level red light with a wavelength of 650 nm and a luminance of 1600 lx through the pupil to the retina. At the retina, the power measures 0.29 mW with a pupil diameter of 4 mm. According to the International Electrotechnical Commission standard, level 1 power is safe for direct eye exposure and does not cause photothermal damage. After the baseline evaluation, participants in the intervention group started the RLRL treatment twice a day, 7 days a week, until the 12-month follow-up, with each session lasting 3 minutes. The participants attended follow-up appointments 1, 3, 6, 9, and 12 months after the baseline evaluation.

Study Outcomes

The primary outcome was the change in axial length (AL) measured at baseline and the 12-month follow-up visit. The secondary outcome was the change in cycloplegic SER measured at baseline and the 12-month follow-up visit. Other ocular biometric parameters included anterior chamber depth (ACD), central corneal thickness (CCT), and lens thickness (LT) measured at baseline compared with that obtained at the 12-month follow-up visits. Safety observations included intraocular pressure (IOP), BCVA, macular OCT, and adverse events reported by the participants and their guardians.

Axial length, ACD, CCT, and LT were measured by a technician using an IOLMaster 700 (Carl Zeiss Meditec), who continued measurements until the standard deviation was <0.05 mm. The IOL Master 700 was measured with cycloplegia at baseline, 6-month, and 12-month visits, without cycloplegia at other visits. The SER was calculated as the sum of the spherical diopter and half of the cylindrical diopter. Technicians measured refraction at least 3 times using an autorefractor (KR8800, Topcon) but performed the measurements until the required specifications were met (sphere difference <0.25 D, cylinder difference <0.25 D, and axis <5°). Cycloplegia was assessed at baseline and the 6- and 12-month follow-ups. Participants received 1% cyclopentolate (Cyclogyl, Alcon) as cycloplegic agent. Intraocular pressure was measured using noncontact tomography (NT-530P, NIDEK). Uncorrected visual acuity (UCVA) and BCVA were assessed by optometrists using the Early Treatment Diabetic Retinopathy Study logarithm of the minimum angle of resolution chart (Guangzhou Xieyi Weishikang) at 4 m from the chart. The macular area was examined using swept-source OCT (SS-OCT; Triton, Topcon).

Intervention Compliance Monitoring

Participants in the intervention group logged into the device with their unique accounts and passwords. The device was networked, emitting low-level red light for 3 minutes twice a day, with at least a 4-hour interval. Use time and date were automatically recorded and sent back information online that managed therapy compliance. The system sends a reminder if a participant failed to undergo treatment twice.

The compliance of treatment was calculated as the percentage of actual number of completed treatments divided by the assigned number of treatments (2 times per day, 7 days per week) for 12 mouths.

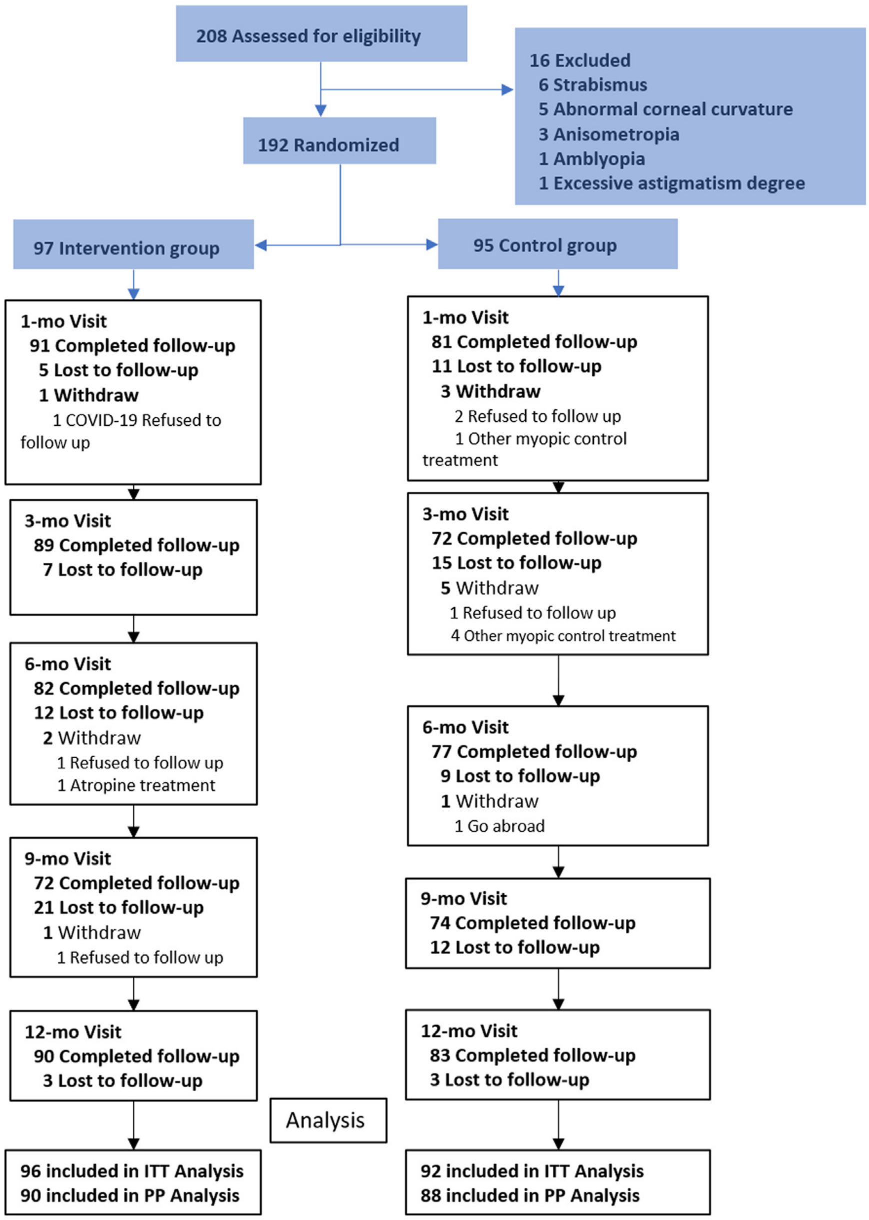

Figure 1. Patient flow in the trial. COVID-19 = Coronavirus disease 2019; ITT = intention to treat; PP = per protocol.

Adverse Effects

All participants were included in the safety analysis. The masked investigator collected and documented any adverse events during each visit. Participants and their guardians reported any adverse effects and were questioned about their experiences related to dizziness, glare, flicker blindness, afterimage time, or vision loss. The potential side effects associated with RLRL were also recorded. A senior ophthalmologist at each study site evaluated the fundus images by OCT to identify underlying lesions without clinical signs.

Sample Size

The sample size estimation was conducted assuming a 2-sided α level of 0.05, 90% power, an annual axial elongation of 0.40 mm with a standard deviation of 0.40 mm over 12 months,[18] and a 50% treatment effect (reducing axial elongation by 0.20 mm). The required sample size was 86 participants per group, totaling 172 participants. After adjusting for a 10% loss to follow-up, the total sample size was 192 participants.

Statistical Analyses

The intention-to-treat population comprised participants randomly assigned to either group. Data from all participants who attended at least 1 subsequent follow-up visit were included in the analysis, regardless of treatment or follow-up visit compliance. No imputation was used for the missing data. Individuals who switched to other myopia treatment modalities, such as orthokeratology or atropine eye drops, or discontinued the RLRL treatment, were censored. If a participant had only 1 eye with an SER exceeding −4.0 D, that eye was selected for statistical analysis. In cases

Table 2. Baseline Characteristics of Included Participants

Characteristics	Intervention (N = 97)	Control (N = 95)
Age, mean (SD), yrs	10.40 (2.4)	11.20 (2.1)
Gender		
Boys, N (%)	63 (65.0)	44 (46.3)
Girls, N (%)	34 (35.0)	51 (53.7)
UCVA, logMAR		
Mean (SD)	0.78 (0.21)	0.75 (0.23)
Median (interquartile range)	0.80 (0.60–1.00)	0.70 (0.50–1.00)
BCVA, logMAR		
Mean (SD)	0.008 (0.037)	0.007 (0.033)
Median (interquartile range)	0 (0–0)	0 (0–0)
BCVA, Snellen equivalent		
< 20/25	2 (2.06)	1 (10.5)
≥ 20/25	95 (97.94)	94 (98.95)
AL, mm		
Mean (SD)	25.93 (1.03)	25.72 (0.87)
Median (interquartile range)	25.86 (25.34–26.48)	25.67 (25.05–26.27)
Spherical equivalent, D		
Mean (SD)	−5.88 (1.69)	−5.75 (1.17)
Median (interquartile range)	−5.50 (−6.56 to −4.69)	−5.44 (−6.38 to −4.88)
IOP, mmHg		
Mean (SD)	17.00 (3.1)	16.90 (2.9)
Median (interquartile range)	17 (15–20)	17 (14–19)
ACD, mm		
Mean (SD)	3.80 (0.24)	3.70 (0.26)
Median (interquartile range)	3.79 (3.64–3.96)	3.71 (3.49–3.87)
LT, mm		
Mean (SD)	3.33 (0.16)	3.37 (0.18)
Median (interquartile range)	3.31 (3.22–3.42)	3.40 (3.23–3.51)
Central thickness, mm		
Mean (SD)	547.72 (31.20)	543.62 (35.50)
Median (interquartile range)	543.01 (526.00–568.00)	545.52 (519.50–564.00)

Data are presented as mean SD, number (%), or median (interquartile range).
ACD = anterior chamber depth; AL = axial length; BCVA = best-corrected visual acuity; D = diopter; IOP = intraocular pressure; logMAR = logarithm of the minimum angle of resolution; LT = lens thickness; SD = standard deviation; UCVA = uncorrected visual acuity.

where both eyes were exceeding −4.0 D, the right eye was chosen for the analysis.

Treatment efficacy was assessed using longitudinal mixed models for primary (changes in AL) and secondary (changes in SER, ACD, CCT, LT, and IOP) outcomes at multiple follow-up visits. The mixed-model repeated-measures analysis with unstructured covariance structures was fit to the individual change from baseline AL (SER/ACD/CCT/LT/IOP), where the group, time, group-by-time interaction, baseline age, sex, and baseline AL (SER/ACD/CCT/LT/IOP) were fixed effects. The primary focus of the analysis was to test treatment differences at 12 months. The main results will consist of the 12-month estimated least square means along with their differences, 95% confidence intervals, and *P* values. The UCVA alterations were classified into 3 categories: a decline by ≥2 lines, stable (within a line), or improvement by ≥2 lines. The BCVA at the 12-month and baseline was evaluated against the 20/25 standard.

Sensitivity analyses were conducted to assess the treatment effects of therapy in controlling myopia progression, specifically AL elongation and progression of myopic SER, across different baseline SER and age groups.

A longitudinal mixed-model analysis was conducted to assess the relationship between treatment efficacy and compliance within the intervention group. Treatment compliance in the intervention group was measured as a percentage of the total number of assigned treatment sessions.

Safety analyses were performed in the intention-to-treat population. All adverse events were reported individually. Six ophthalmologists (Y.X., H.Z., H.P.C., X.L.F., K.H.F., and C.P.S.), 1 from each center, independently scrutinized the OCT scans for potential structural anomalies.

P values for statistical significance in all analyses were based on 2-sided tests, with a significance level of $P < 0.05$. Statistical analyses were conducted using SAS version 9.4 (SAS Institute, Inc).

Results

Between February 6, 2021, and April 15, 2022, 208 participants were screened at 6 centers, and 192 (92.31%) were enrolled; 97 and 95 participants were randomly assigned to the intervention and control groups, respectively. Recruitment stopped once the required sample size was reached. Table S1 (available at www.aaojournal.org) shows the number of participants enrolled in each center, and Figure 1 presents the number of participants who completed the screening, enrollment, and follow-up processes at each time point.

Because of the influence of Coronavirus disease 2019 and related restrictions, some participants did not complete all follow-ups. In total, 173 participants (90.1%) completed the 12-month follow-up period; 90 (92.78%) were from the

Table 3. Results of Efficacy Assessment at 12-Month Follow-up

Variables	Intervention (N = 96)	Control (N = 92)	Difference (95% CI)	P Value
Primary outcome				
Mean change in AL from baseline, mm	−0.060 (−0.100 to −0.020)	0.340 (0.300−0.390)	0.410 (0.350−0.470)	<0.0001
Secondary outcomes				
Mean change in spherical equivalent from baseline, D	0.100 (−0.020 to 0.220)	−0.75 (−0.880 to −0.62)	−0.860 (−1.040 to −0.680)	<0.0001
Mean change in IOP from baseline, mmHg	−0.290 (−0.870 to 0.300)	−0.56 (−1.160 to 0.040)	−0.270 (−1.110 to 0.570)	0.427

AL = axial length; CI = confidence interval; D = diopter; IOP = intraocular pressure.

intervention group, and 83 (87.37%) were from the control group. Overall, 188 participants were included in the statistical analysis: 96 from the intervention group and 92 from the control group. One participant from the intervention group and 3 participants from the control group were excluded because of a lack of visits after the baseline examination. Four participants from the intervention group ceased RLRL treatment and dropped out of the trial; 3 participants withdrew because of an unwillingness to continue follow-up, and 1 participant switched to atropine drops. Nine participants from the control group withdrew from the trial; 1 was unable to attend the follow-up appointments because of being abroad, 3 were unwilling to attend, and 5 opted for alternative myopia treatments. The final visits of these participants before withdrawing were analyzed.

Baseline Characteristics

The mean ages of the participants in the intervention and control groups were 10.4 ± 2.4 years (63 male participants [65.0%]) and 11.2 ± 2.1 years (44 male participants [46.3%]). The baseline UCVA, BCVA, AL, SER, and IOP values did not differ between the intervention and control groups (Table 2).

Primary Outcome

The adjusted mean AL change at 12 months was −0.06 mm (95% CI, −0.10 to −0.02 mm) in the intervention group and 0.34 mm (95% CI, 0.30−0.39 mm) in the control group. Thus, the adjusted mean difference in AL change between the 2 groups at the 12-month follow-up was 0.41 mm (95% CI, 0.35−0.47 mm; $P < 0.0001$; Table 3, Figure 2, and Table S4 [available at www.aaojournal.org]). Figure 2 and Table S4 present the adjusted mean differences in AL between the intervention and control groups at the 1-, 3-, 6-, and 9-month follow-ups. The mixed model analysis of the primary outcome showed that age, group, time, and the interaction between group and time were significant variables (Table S5, available at www.aaojournal.org).

Notably, the adjusted mean change in AL for the intervention group was negative at all follow-up assessments and exceeded the IOLMaster instrument's margin of error by >0.05 mm, indicating axial shortening. During the 12-month follow-up, 48 participants (53.3%) of the intervention group were still experiencing axial shortening. Table S6 (available at www.aaojournal.org) presents the proportion of participants with >0.05 mm of axial shortening in both groups at each follow-up visit.

Secondary Outcomes

The mean SER change after 12 months was 0.11 D (95% CI, 0.02−0.19) in the intervention group and −0.75 D (95% CI, −0.88 to −0.62) in the control group. The mean SER difference between the intervention and control groups after 12 months was −0.86 D (95% CI, −1.04 to −0.68; $P <$

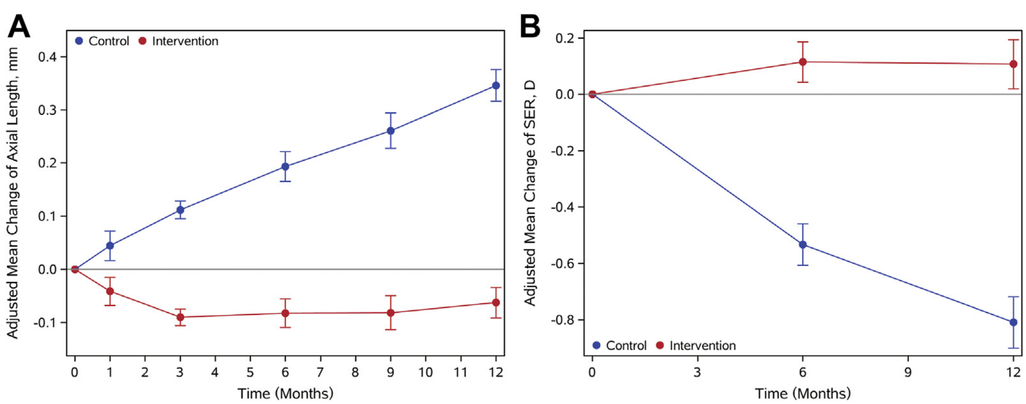

Figure 2. Axial length (AL) elongation and spherical equivalent myopic shifts over 12 months. Error bars represent standard deviations. D = diopters; SER = spherical equivalent refraction.

Table 8. Changes in Uncorrected Visual Acuity and Best-Corrected Visual Acuity from Baseline to 12 Months between the Repeated Low-Level Red Light Group and Control Group

	Intervention	Control	Difference (95% CI)	
Change in UCVA				
2 lines worsening	10 (14.7%)	20 (31.8%)	−17.1 (−31.3% to −2.8%)	0.047
Within 1 line	46 (67.6%)	37 (58.7%)	8.9 (−7.6% to 23.4%)	
2 lines improvement	12 (17.6%)	6 (9.5%)	8.1 (−3.5% to 19.7%)	
BCVA				
No. <20/25	2/87 (2.3)	1/83 (1.2%)	1.1 (−2.8% to 5.0%)	

BCVA = best-corrected visual acuity; CI = confidence interval; UCVA = uncorrected visual acuity.

0.0001; Tables 3 and S4, available at www.aaojournal.org, and Figure 2). Table S4 (available at www.aaojournal.org) and Figure 2 present the mean SER difference after 6 months. The mixed model of SER identified group, time, and the interaction between the 2 as significant variables (Table S5, available at www.aaojournal.org). Positive SER changes occurred at the 6- and 12-month follow-ups in the intervention group, indicating refraction regression. All values were <0.25 D and did not exceed the margin of error. The intervention group was similar to the control group for mean changes in other ocular biometric parameters (ACD, CCT, and LT). The 1-, 3-, 6-, 9-, and 12-month adjusted mean changes of these ocular parameters for each group and mean differences between the intervention and control groups are presented in Table S7 (available at www.aaojournal.org).

At the 12-month follow-up, more participants in the intervention group had a 2-line UCVA improvement than in the control group (12 participants vs. 6 participants, Table 8). The number of participants with a BCVA of <20/25 remained the same as that at baseline in both groups. The IOP did not differ between the 2 groups, and structural damage was not detected on the photosensory layer of macular SS-OCT during the trial in any participant.

Compliance

The intervention group demonstrated an 84% median RLRL treatment compliance rate (interquartile range, 72%−92%). Table S9 (available at www.aaojournal.org) presents the correlations between compliance and changes in AL and SER, which were statistically insignificant.

Adverse Effects

No severe adverse events, including sudden vision loss by 2 lines or scotoma, developed during the trial. Following meticulous inquiries by the clinical researchers, no adverse effects such as glare, flickering, or afterimages lasting longer than 5 minutes were reported. Additionally, careful examinations conducted by senior ophthalmologists revealed no signs of structural damage, including but not limited to tissue disruption, discontinuity, edema, or abnormalities in the photosensory layer, as detected by SS-OCT imaging. However, 1 participant in the intervention group reported conjunctivitis during the RLRL treatment.

Sensitivity and Subgroup Analyses

Table S10 (available at www.aaojournal.org) presents the sensitivity analysis results, which were similar to the primary findings, verifying the robustness of the main findings of this study. Subgroup analyses were conducted to compare the myopia control effect (including AL and SER changes) in participants with varying baseline SER values (Table S11, available at www.aaojournal.org) and in different age groups (Table S12, available at www.aaojournal.org). The treatment effect (AL change, $P = 0.207$; SER change, $P = 0.182$) did not vary significantly by baseline SER (−4.0 D to −6.0 D vs. <−6.0 D). In terms of age, participants aged 12 to 16 years exhibited less AL elongation compared with those aged 6 to 11 years ($P = 0.044$). Additionally, the change in SER was not significantly different between these age groups ($P = 0.589$). Figure S3 (available at www.aaojournal.org) shows the distribution of SER ranges among subjects in the intervention group and the control group.

Discussion

This study is among the earliest clinical trials investigating myopia control among children and adolescents with high myopia. Repeated low-level red light (RLRL) treatment slowed AL elongation by 0.41 mm and progression of myopic SER by 0.86 D compared with SVS. However, after 12 months, the adjusted mean AL change of the intervention group still had negative values, and 53.3% of them with >0.05 mm of axial shortening. Overall, our study suggests that RLRL treatment is an excellent solution for obtaining myopia control in children with high myopia, and it is the only existing way to shorten the AL.

Axial Length Shortening

Traditionally, myopia has been regarded as an irreversible condition characterized by progressive axial elongation.[1,2] High myopia is characterized by excessive elongation in the AL, resulting in substantial visual impairments.[19] Before the introduction of RLRL therapy, no human studies had demonstrated the possibility of any myopia control treatment causing axial shortening over months or years. However, Jiang et al[13] and Wang et al[20] was the first to report axial shortening in myopic children undergoing RLRL treatment. Among the children with myopia of SER of −1.0 D to −5.0 D who received RLRL treatment, the adjusted mean AL change was negative at both 1-month and 3-month intervals. Interestingly, after 12 months, 21.6% of the subjects in the RLRL group exhibited an AL shortening of >0.05 mm, even though the adjusted mean AL change had become positive by then. In our study involving children with high myopia, the RLRL group consistently showed a negative adjusted mean AL change during each follow-up. Of note, after 12 months, 53.3% of participants in the intervention group showed an AL shortening exceeding 0.05 mm.

The most relevant measure to the development of pathological myopia is the length from the cornea to the sclera, which indicates abnormal scleral expansion. However, refraction involves the plane of the photoreceptor outer segments. If the total axial elongation (cornea to sclera) is less than the expansion of the choroid, then methods of measurement such as the IOLMaster, which measure to the inner limiting membrane, will report reductions in AL that do not necessarily reflect changes in the cornea to sclera measure of AL. Nonetheless, the measurement of the cornea to scleral length remains elusive, and a previous study by Xiong et al[17] and Du et al[19] documenting a RLRL group that exhibited choroidal thickness thickening (0.056 mm) could only explain 28.3% of AL shortening (−0.20 mm). On analyzing the choroidal data of a subset of participants in this study, we found that after 12 months of RLRL treatment, children in the intervention group experienced an average increase in choroidal thickness of 0.022 mm (Table S13, available at www.aaojournal.org). However, this increase cannot fully account for the AL shortening observed at the end of the trial. This suggests that the AL shortening, which was also observed in this study, may be due to potential scleral remodeling or contraction. Moreover, to address the challenges associated with measuring AL because of variations in choroidal thickness, it is essential to discuss strategies to mitigate these measurement discrepancies. Implementing washout periods and conducting systematic OCT measurements of choroidal thickness could be effective approaches. Addressing these methodological issues is crucial because they potentially affect the reliability of all myopia control interventions.

Repeated Low-Level Red Light Versus Other Myopia Control Treatments

To date, there have been only a few randomized controlled studies specifically reporting the efficacy of myopia control strategies among highly myopic children. At present, the most widely recognized and practiced approach to myopia control is atropine administration.[6] However, neither the renowned ATOM nor LAMP studies included participants with a SER exceeding −6.0 D. Children with higher myopia who use the orthokeratology lens need an extra frame glass to maintain good daytime vision.

A high-myopia subgroup from an orthokeratology study[21] demonstrated a mean AL elongation of 0.16 mm within the initial 12 months of intervention. It is clear that the RLRL treatment has obvious advantages in controlling AL increase.

Repeated Low-Level Red Light Effects on Different Baseline Spherical Equivalent Refraction

In the study by Jiang et al[13] on children with a SER of −1.0 D to −5.0 D, the group receiving RLRL treatment displayed a higher annual elongation of AL and progression of myopic SER (0.13 mm, −0.2 D) compared with the RLRL group in the current research (−0.06 mm, 0.1 D). In the study by Jiang et al,[13] it was discovered that RLRL has a dose-dependent effect on controlling myopia progression. The study reported that subjects with better compliance had a slower elongation in AL and progression of myopic SER. Given that all subjects in this study were young children with SERs <−4.0 D and the necessity to slow down myopia progression was urgent, we developed a treatment of 7 days a week, while the other parameters remained the same. The higher compliance (with a median of 84% in this study and 75% in Jiang et al) and the increased use of RLRL may account for the lower annual elongation of AL and progression of myopic SER in the intervention group of this study. This factor may explain why the intervention group performed better in our study than in the study by Jiang et al.[13] Additionally, it is important to consider that the severity of myopia may correlate with the extent of scleral hypoxia and inflammation.[22] Consequently, eyes with higher degrees of myopia might experience more significant improvements when subjected to RLRL. This suggests that the therapeutic efficacy of red light could be proportionally greater in children with a higher degree of myopia.

On conducting subgroup analyses, we noted that participants, both with an initial SER between −4.0 and −6.0 D and that exceeding −6.0 D, demonstrated remarkable myopia control efficacy. The adjusted mean changes in AL for the RLRL group were negative values in both subgroups. For individuals seeking effective myopia management strategies, RLRL appears to be a promising alternative.

Safety

As research progresses in the field of RLRL for myopia management, it is imperative to meticulously monitor treatment safety. The application of RLRL therapy, which uses laser diodes, is classified under the category of class 1 light according to the International Electrotechnical Commission's standard 60825-1:2014. This designation implies that the light is deemed safe for direct ocular exposure, delivering a restricted quantity of energy to the eye. Nonetheless, it is important to note that these classifications predominantly pertain to accidental, rather than intentional, repeated exposures.[23] Furthermore, although the Eyerising device exhibits a power output of 0.29 mW, the actual energy level reaching the retina might be substantially augmented after traversing through a pupil of 4-mm diameter. Regrettably, there is a lack of data concerning the precise power levels at the retina when using the Eyerising device, highlighting a gap in current research.

High myopia has weak fundus structure for its decreased retinal and choroid thickness, putting them at a higher risk of fundus damage.[24] Encouragingly, during the follow-up period, no signs of fundus structural damage were detected from the OCT images of intervention group and no other adverse responses were observed. One child was diagnosed with conjunctivitis during the follow-up period which was attributed to an infection or allergy and, subsequently, recovered with attention to eye hygiene. However, we must not become complacent. It may be that the transient choroidal thickening coupled with AL shortening observed in low-level red light therapy indicates possible inflammation or phototoxic damage incurred by repeated exposure to laser-like red light.[23] Liu et al[25] reported a case where a child with high myopia experienced a decline in vision accompanied by a disruption in the ellipsoid zone after 5 months of RLRL use. Although the child's vision recovered after discontinuing the use, and the parents reported a history of prolonged afterimage duration (exceeding 8 minutes) without cessation before the onset of vision decline, this case underscores the importance of rigorous monitoring for adverse effects.

Therefore, in addition to observing the changes in anatomic structure, the changes in early retinal functions (e.g., multifocal electroretinogram responses) should be considered. The treatment safety of red light therapy should be assessed from both physiological and substructural levels.

Limitations

This study had several limitations. First, it is common to use −6 D (or less commonly −5 D) as the cutoff for high myopia. A different cutoff was used in this study, −4 D, because higher levels are rare in children, and there appears to be a significant increase in the prevalence of conventionally defined high myopia only after the age of approximately 11 to 12 years. Although children aged 6 to 16 years with myopia worse than −4 D can be considered highly myopic, this classification does not align with the conventional definition of high myopia. Another limitation of this study is that the collection of choroidal thickness data was not initially planned. Unfortunately, because of data loss for many participants during the Coronavirus disease 2019 pandemic, we can only provide choroidal data for a small subset of participants. This limitation is regrettable and highlights the importance of proactively incorporating choroidal thickness and washout period into future myopia study designs. Second, this study used RLRL therapy for controlling high myopia development for only 1 year, and the evaluation of its long-term efficiency and safety requires further clinical research for validation. Furthermore, neither the control group nor the intervention group in this study was masked, which may introduce certain biases in the research results. Last, because of competitive enrollment across the study's various centers, there was significant variation in the number of participants recruited by each of the 6 centers. Although adjustments have been made to the analysis to account for this discrepancy, it remains a limitation of the study. Future studies should address these limitations.

Repeated low-level red light demonstrates much stronger treatment efficacy among high myopia in children and adolescents aged 6 to 16 years, with 53.3% experiencing substantial axial shortening. Repeated low-level red light provides an excellent solution for the management of high myopia progression, a significant challenge in ophthalmology practice. It is necessary to conduct further clinical research to determine whether AL shortening and SER regression observed after 1 year of RLRL therapy are temporary or persistent phenomena as well as whether the treatment effects can be maintained after discontinuing red light therapy.

Footnotes and Disclosures

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Analysis and interpretation: Xu, X. Feng, K. Feng, Zhu, H. Cui, Shi, Miao, Li, Zhang, L. Cui

Obtained funding: Xu

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Abbreviations and Acronyms:

ACD = anterior chamber depth; **AL** = axial length; **BCVA** = best-corrected visual acuity; **CCT** = central corneal thickness; **CI** = confidence interval; **D** = diopters; **IOP** = intraocular pressure; **LT** = lens thickness; **RLRL** = repeated low-level red light; **SER** = spherical equivalent refraction; **SS-OCT** = swept-source OCT; **SVS** = single-vision spectacles; **UCVA** = uncorrected visual acuity.

Key words:

Axial shortening, Children and adolescents, High myopia, Repeated low-level red light.

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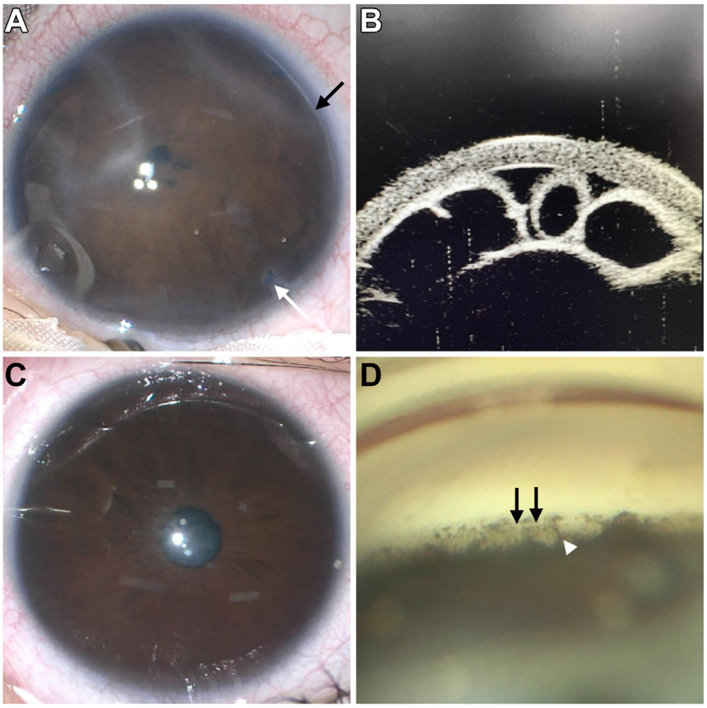
Pictures & Perspectives



An Iris Cyst as a Presenting Feature of Axenfeld-Rieger Malformation

A 3-month-old girl presented with a whitish opacity in her right eye (OD). Examination revealed an abnormal iris morphology OD. She had a posterior embryotoxon (PE; **A,** black arrow), along with peripheral iris atrophy and holes (**A,** white arrow). On ultrasound biomicroscopy, multiple loculated cysts could be observed OD (**B**). Her left eye had a grossly normal iris morphology (**C**), although on gonioscopy a well-defined PE (**D**, black arrows), with high iris processes were observed (**D**, white arrowhead). The child had a known mutation (p.Gly34ThrfsTer8) that is associated with Axenfeld-Rieger malformation. (Magnified version of Figure **A**-**D** is available online at www.aaojournal.org).

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